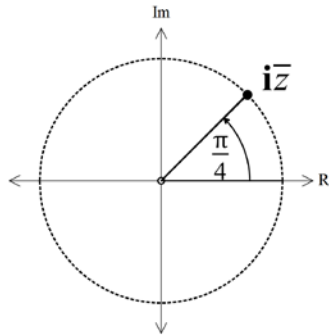


Section 1: Multiple Choice (10 marks)

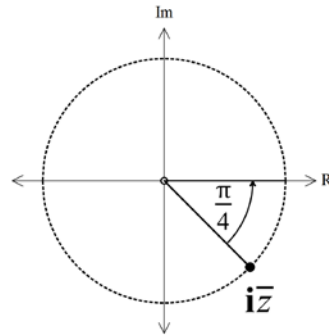
Indicate your answers on the answer sheet provided.

Q1. Given the complex number z has $\text{Arg } z = \frac{\pi}{4}$, which of the following is a correct representation of $i\bar{z}$?

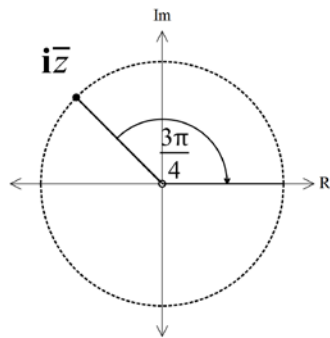
A



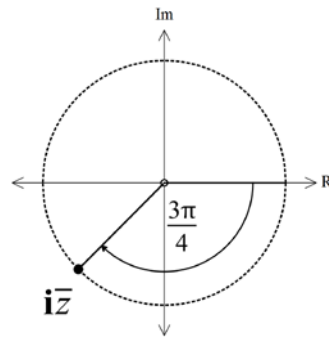
B



C



D



Q2. Which expression is equal to $\int_0^a [f(a-x) + f(a+x)]dx$?

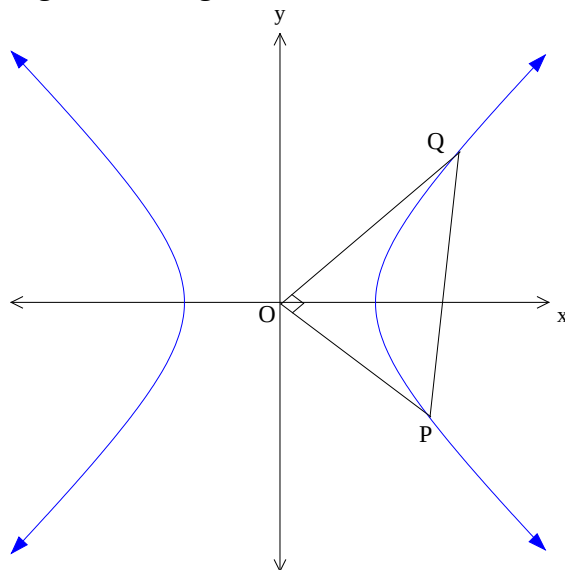
A $\int_0^a f(x)dx$

B $\int_0^{2a} f(x)dx$

C $2 \int_0^a f(x)dx$

D $\int_{-a}^a f(x)dx$

- Q3. The diagram below shows the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ where $a > b > 0$. The points $(a \sec \theta, b \tan \theta)$ and $Q (a \sec \alpha, b \tan \alpha)$ lie on the hyperbola and the chord PQ subtends a right angle at the origin.



Use the parametric representation of the hyperbola to determine which of the following expressions is correct?

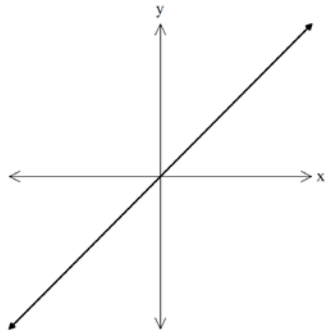
- A $\sin \theta \sin \alpha = -\frac{a^2}{b^2}$
- B $\sin \theta \sin \alpha = \frac{a^2}{b^2}$
- C $\tan \theta \tan \alpha = -\frac{a^2}{b^2}$
- D $\tan \theta \tan \alpha = \frac{a^2}{b^2}$
- Q4. The polynomial equation $x^3 + 3x^2 - 2x + 6 = 0$ has roots α, β and γ .

Which of the following polynomials has roots $\alpha - 1, \beta - 1, \gamma - 1$?

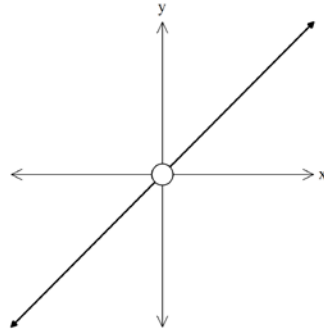
- A $x^3 - 5x + 10 = 0$
- B $x^3 + 3x^2 - 2x + 14 = 0$
- C $x^3 + 6x^2 + x + 8 = 0$
- D $x^3 + 6x^2 + 7x + 8 = 0$

Q5. Which of the following graphs best represents the implicit function $\frac{x}{y} + \frac{y}{x} = 2$?

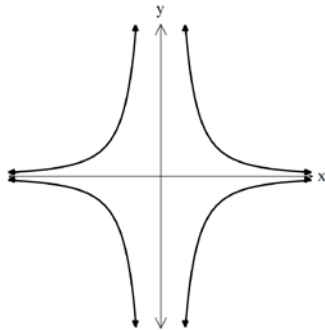
A



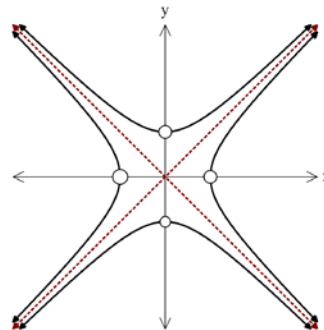
B



C



D



Q6. Evaluate $\int_0^{\frac{\pi}{2}} \frac{1}{\cos\theta + 2\sin\theta + 3} d\theta$ using the substitution $t = \tan\frac{\theta}{2}$

A 0.322

B 0.785

C 1.107

D 1.570

Q7. The point $P(z)$ moves on the complex plane according to the condition

$|z - i| + |z + i| = 4$. The Cartesian equation of the locus of P is:

A $\frac{x^2}{4} + \frac{y^2}{3} = 1$

B $\frac{x^2}{3} + \frac{y^2}{4} = 1$

C $x^2 + y^2 = 1$

D $x^2 + y^2 = 4$

Q8. The region bounded by $y \leq 4x^2 - x^4$ and $0 \leq x \leq 2$ is rotated about the y axis to form a solid. What is the volume of this solid using the method of cylindrical shells?

A $\frac{16\pi}{3} \text{ units}^3$

B $\frac{8\pi}{3} \text{ units}^3$

C $\frac{20\pi}{3} \text{ units}^3$

D $\frac{32\pi}{3} \text{ units}^3$

Q9. What is the area of the largest rectangle that can be inscribed in the ellipse

$4x^2 + 9y^2 = 36$?

A $24\sqrt{2}$

B $6\sqrt{2}$

C 24

D 12

Q10. Consider the equation $\left(\frac{2+i}{c}\right)^p = 1$ where c is a real and $p \neq 0$

For how many values of c will this equation have real solutions?

- A None
- B One
- C Two
- D Four

End of Multiple Choice

Section II (90 marks)

Attempt Questions 11–16

Allow about 2 hours and 45 minutes for this section

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available. In Questions 11–16, your responses should include relevant mathematical reasoning and/or calculations.

Question 11

Use a SEPARATE writing booklet.

(15 marks)

(a) Find $\int \tan^4 x \sec^2 x dx$ (1)

(b) Use the substitution $u = x - 2$ to evaluate $\int_{\frac{3}{2}}^{\frac{5}{2}} \frac{1}{\sqrt{(x-1)(3-x)}} dx$ (3)

(c) (i) Write $\frac{3x+2}{x^2+5x+6}$ as a sum of partial fractions. (2)

(ii) Hence evaluate $\int_0^2 \frac{3x+2}{x^2+5x+6} dx$ (2)

(d) Given that the point $P(6\cos\theta, 2\sin\theta)$ lies on an ellipse, determine the following:

(i) The eccentricity. (1)

(ii) Coordinates of the foci. (1)

(iii) Equations to the directrices. (1)

(iv) Sketch neatly this ellipse showing all important features. (1)

(e) For the hyperbola $\frac{x^2}{16} - \frac{y^2}{25} = 1$

(i) Find the eccentricity. (1)

(ii) Find the equations of the asymptotes. (1)

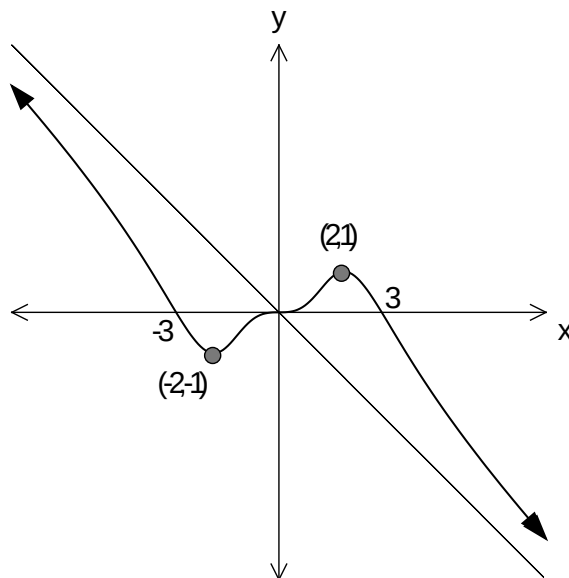
(iii) If P is on the hyperbola and S and S' are its foci, then given PS=2, find PS'. (1)

Question 12

Use a SEPARATE writing booklet.

(15 marks)

- (a) Realise the denominator of $\frac{7 - 2i}{3 + i}$ (1)
- (b) (i) On the same diagram, sketch the locus of both $|z - 2| = 2$ and $|z| = |z - 4i|$. (2)
- (ii) What is the complex number represented by the point of intersection of these two loci? (1)
- (c) Let z be a complex number of modulus 3 and ω be a complex number of modulus 1. Show that $|z - \omega|^2 = 10 - (\overline{z\omega} + \overline{z}\omega)$ (2)
- (d) Given the polynomial $2x^3 + 3x^2 - x + 1 = 0$ has roots α, β and γ :
- (i) Find the polynomial whose roots are α^2, β^2 and γ^2 . (2)
- (ii) Determine the value of $\alpha^3 + \beta^3 + \gamma^3$. (2)
- (e) The graph of $y = f(x)$ is shown. The line $y = -x$ is an oblique asymptote to the curve.



Use separate one-third page graphs, to sketch:

- (i) $f(-x)$ (1)
- (ii) $f(|x|)$ (1)
- (iii) $\frac{x}{f(x)}$ (3)

Question 13

Use a SEPARATE writing booklet.

(15 marks)

(a) Given that $1 + i$ is a zero of $P(x) = 0$ where $P(x) = x^4 - x^3 - 2x^2 + 6x - 4$, factorise $P(x)$ fully over the field of the complex numbers. (2)

(b) The ellipse $\frac{(x-4)^2}{9} + \frac{y^2}{4} = 1$ is rotated about the y axis to form a solid of revolution.

(i) By taking slices perpendicular to the axis of rotation, show that the volume of a slice is $8\pi\sqrt{36 - 9y^2} \delta y$ (2)

(ii) Find the exact volume of the solid. (2)

(c) Find all the roots of the equation $18x^3 + 3x^2 - 28x + 12 = 0$, given that two of the roots are equal. (3)

(d) P and Q are two points on the same branch of the rectangular hyperbola $xy = 25$. Given that P is $(5p, \frac{5}{p})$ and Q is $(5q, \frac{5}{q})$:

(i) Show that PQ has the equation $x + pqy = 5(p+q)$ where P and Q are parameters p and q respectively. (2)

(ii) If PQ has a constant length of m^2 , show that

$$25[(p+q)^2 - 4pq] (p^2q^2 + 1) = m^4p^2q^2 \quad (2)$$

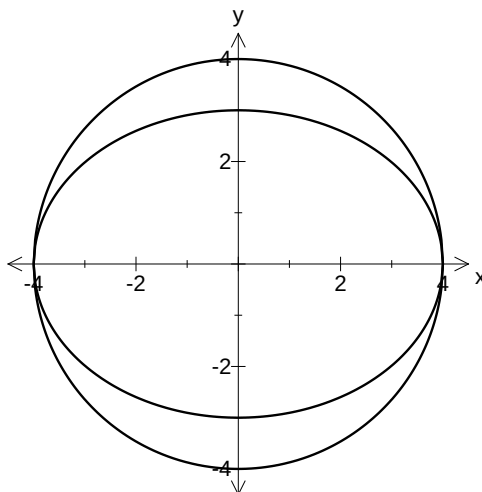
(iii) Show that the locus of R, the midpoint of PQ, in Cartesian form is $xym^4 = 4(xy - 25)(x^2 + y^2)$. (2)

Question 14

Use a SEPARATE writing booklet.

(15 marks)

- (a) The diagram shows an ellipse with equation $\frac{x^2}{16} + \frac{y^2}{9} = 1$ and the larger of its auxiliary circles. The coordinates of a point P on the ellipse are $(4\cos \theta, 3\sin \theta)$ where $\theta \neq 0$ or π .



A straight line l parallel to the y axis intersects the x axis at N and the ellipse and the auxiliary circle at the points P and Q respectively.

- (i) Find the equations of the tangent to the ellipse at P and to the auxiliary circle at Q. (4)
 - (ii) The tangents at P and Q intersect at point R. Show that R lies on the x axis. (2)
 - (iii) Prove that $ON \times OR$ is independent of the positions of P and Q. (1)
- (b) By expanding $(\cos \theta + i\sin \theta)^3$ it can be shown that $\cot 3\theta = \frac{t^3 - 3t}{3t^2 - 1}$ where $t = \cot \theta$.
(Do NOT prove this.)
- (i) Solve $\cot 3\theta = -1$ for $0 \leq \theta \leq 2\pi$ (2)
 - (ii) Hence show that $\cot \frac{\pi}{12} \cdot \cot \frac{5\pi}{12} \cdot \cot \frac{9\pi}{12} = -1$ (2)
 - (iii) Write down a cubic equation with roots $\tan \frac{\pi}{12}, \tan \frac{5\pi}{12}$ and $\tan \frac{9\pi}{12}$ (1)
(Express your answer as a polynomial equation with positive integer coefficients).
- (c) (i) Show that the derivative of the function $y = x^x$ for $x > 0$ is $(\ln x + 1)x^x$. (1)
- (ii) Hence or otherwise neatly draw $y = x^x$ for $x > 0$ (2)

Question 15

Use a SEPARATE writing booklet.

(15 marks)

- (a) A triangle ABC is right-angled at A and it has sides of lengths a , b and c units. A circle of radius r units is drawn so that the sides of the triangle are tangents to the inscribed circle.

Prove that $r = \frac{1}{2}(c + b - a)$. (3)

- (b) (i) A solid has as its base the region bounded by the curves $y = x$ and $x = 2y - \frac{y^2}{2}$.
Cross sections parallel to the x axis are equilateral triangles with a side in the base.

Show that the volume is given by $V = \frac{\sqrt{3}}{4} \int_0^2 (y^2 + \frac{y^4}{4} - y^3) dy$ (3)

(ii) Calculate the volume of this solid. (2)

(c) (i) If $I_n = \int_0^1 (1 - x^2)^{\frac{n}{2}} dx$ where n is a positive integer, show that (3)

$$I_n = \frac{n}{(n+1)} I_{n-2}.$$

(ii) Hence evaluate I_5 . (2)

- (d) Find the exact range of values for m , where m is a non-negative real number, such that $x^2 + (2 - m^2)y^2 = 1$ represents an ellipse in the Cartesian Plane. (2)

PTO for Q16

Question 16

Use a SEPARATE writing booklet.

(15 marks)

(a) (i) Show that $a^2 + b^2 \geq 2ab$ where a and b are distinct positive real numbers. (1)

(ii) Hence show that $a^2 + b^2 + c^2 \geq ab + ac + bc$. (1)

(iii) Hence show that $\sin^2\alpha + \cos^2\alpha \geq \sin 2\alpha$. (1)

(iv) Hence show that $\sin^2\alpha + \cos^2\alpha + \tan^2\alpha \geq \sin \alpha - \cos \alpha + \sec \alpha + \frac{1}{2} \sin 2\alpha$. (1)

(b) An orchestra has $2n$ cellists; n being female and n male. From the $2n$ cellists a committee of three members is formed which contains more females than males. Two members of the orchestra cellists are Paul and Matilda. Find the probability that a committee chosen at random has Matilda in it if it is known that Paul has been chosen. (1)

(c) When a polynomial $P(x)$ is divided by $x^2 - a^2$, where $a \neq 0$, the remainder is $px + q$.

(i) Show that $p = \frac{1}{2a} [P(a) - P(-a)]$ and $q = \frac{1}{2} [P(a) + P(-a)]$ (2)

(ii) Find the remainder when $P(x) = x^n - a^n$ is divided by $x^2 - a^2$ and n is a positive integer. (2)

(d) For $n = 1, 2, 3, \dots$ let $S_n = 1 + \sum_{r=1}^n \frac{1}{r!}$.

(i) Prove by Mathematical Induction that $e - S_n = \int_0^1 \frac{x^n}{n!} e^{-x} dx$ (3)

(ii) Deduce that $0 < e - S_n < \frac{3}{(n+1)!}$ for $n = 1, 2, 3, \dots$ (1)

(iii) Prove that $(e - S_n)n!$ is not an integer for $n = 2, 3, 4, \dots$ (1)

(iv) Show that there cannot exist positive integers p and q such that $e = \frac{p}{q}$. (1)

END OF EXAM

Multiple Choice.

1. A
2. B
3. A
4. D
5. B
6. A
7. B
8. D
9. D
10. C

conjugate of z followed by 90° anticlockwise rotation

$$\frac{x^2}{9} - \frac{y^2}{4} = 1$$

$$e^2 = \frac{13}{4} \quad e = \frac{\sqrt{13}}{2}$$

$$\textcircled{5} \quad x^2 + y^2 = 2xy \quad (x-y)^2 = 0 \quad y = x \quad x \neq 0, y \neq 0$$

$$y = x-1 \quad \therefore x = y+1 \quad (y+1)^3 + 3(y+1)^2 - 2(y+1) + 6 =$$

$$\delta V = \pi (R^2 - r^2) \delta x = \pi [1 - \sin^2 x] dx$$

$$\frac{x^2}{9} + \frac{y^2}{4} = 1 \quad \text{Area} = (2x)(2y) = 4xy$$

$$\left| \frac{2+i}{c} \right| = 1$$

$$|c| = |2+i| = \sqrt{5}$$

$$\text{since } c \in \mathbb{R}, c = \pm \sqrt{5}$$

Question 11

$$\textcircled{a} \quad \int \tan^4 x \sec^2 x dx = \frac{1}{5} \tan^5 x + C \quad \textcircled{1}$$

$$\textcircled{b} \quad u = x-2 \quad \text{when } x = 3/2, u = -1/2$$

$$u+1 = x-1 \quad x = 5/2, u = 1/2$$

$$1-u = 3-x$$

$$\therefore \int_{3/2}^{5/2} \frac{1}{\sqrt{(x-1)(3-x)}} dx \Rightarrow \int_{-1/2}^{1/2} \frac{1}{\sqrt{(1+u)(1-u)}} du \quad \textcircled{1}$$

$$= \int_{-1/2}^{1/2} \frac{1}{\sqrt{1-u^2}} du$$

$$= [\sin^{-1} u]_{-1/2}^{1/2} \quad \textcircled{1}$$

$$= \sin^{-1}(1/2) - \sin^{-1}(-1/2)$$

$$= \frac{\pi}{6} - \left(-\frac{\pi}{6}\right) = \frac{\pi}{3} \quad \textcircled{1}$$

$$(c) (i) \frac{3x+2}{(x+3)(x+2)} \equiv \frac{A}{x+3} + \frac{B}{x+2}$$

$$3x+2 = A(x+2) + B(x+3)$$

$$\text{when } x = -2; \quad -4 = B$$

$$\text{when } x = -3; \quad -7 = -A \therefore A = 7 \quad \} \quad (1)$$

$$\therefore \frac{+7}{x+3} - \frac{4}{x+2}$$

(1)

$$(ii) \int_0^2 \frac{3x+2}{(x+3)(x+2)} dx = \int_0^2 \left(\frac{7}{x+3} - \frac{4}{x+2} \right) dx$$

$$= \left[7 \ln(x+3) - 4 \ln(x+2) \right]_0^2 \quad (1)$$

$$= 7 \ln 5 - 4 \ln 4 - 7 \ln 3 + 4 \ln 2$$

$$= 7 \ln\left(\frac{5}{3}\right) - 4 \ln(2) \quad (1)$$

$$\text{or} // = \ln\left(\frac{78125}{34992}\right)$$

$$d) (i) P(6 \cos \theta, 2 \sin \theta) \therefore \frac{x^2}{36} + \frac{y^2}{4} = 1$$

$$b^2 = a^2(1 - e^2)$$

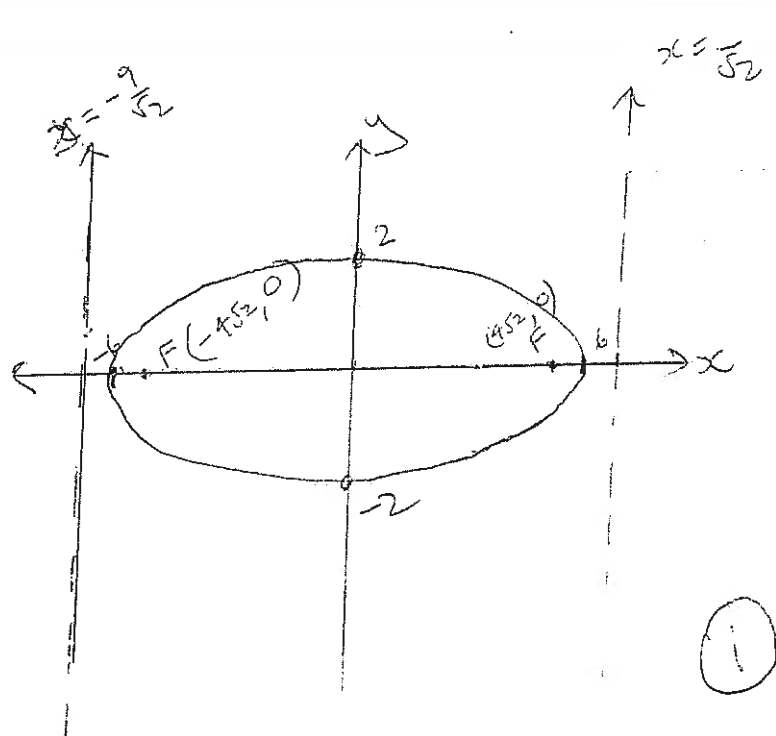
$$e = \sqrt{1 - \frac{4}{36}} = \sqrt{\frac{32}{36}}$$

$$\therefore e = \frac{2\sqrt{2}}{3} \quad (1)$$

$$(ii) \text{ Foci: } (\pm ae, 0) \\ = (\pm 4\sqrt{2}, 0) \quad (1)$$

$$(iii) \text{ Directrices is } x = \pm \frac{a}{e} = \pm \frac{6}{2\sqrt{2}} = \pm \frac{3\sqrt{2}}{2} \quad (1)$$

(iv)



(e) (i) $\frac{x^2}{16} - \frac{y^2}{25} = 1$
 $b^2 = a^2(e^2 - 1)$
 $25 = 16(e^2 - 1)$

$$e^2 = 1 + \frac{25}{16}$$

$$e = \pm \frac{\sqrt{41}}{4}$$

but $e > 0$ for hyperbola

$$\therefore e = \frac{\sqrt{41}}{4} \text{ only}$$

(1)

(ii) $y = \pm \frac{b}{a}x$ $\therefore y = \pm \frac{5}{4}x$

(1)

(iii) $|PS - PS'| = 2a$
 $|2 - PS'| = 8$

(as $a = 4$)

$$PS' = 10 \text{ or } -6$$

but $PS' > 0$ (distance)

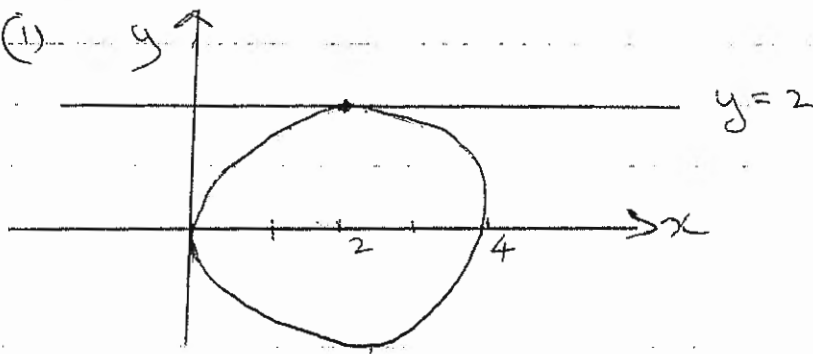
$$\therefore PS' = 10 \text{ only}$$

(1)

Question 12

$$(a) \frac{7-2i}{3+i} \times \frac{3-i}{3-i} = \frac{21-6i-7i-2}{9+1} = \frac{19-13i}{10} \quad (1)$$

(b) (i)



(1) line
(1) circle.

(ii) line and circle intersect at $2+2i$ (1)

(*) as $z\bar{z} = |z|^2 \therefore (z-w)^2 = (z-w)\overline{(z-w)}$

$$= (z-w)(\bar{z}-\bar{w}) \quad (1)$$

$$= z\bar{z} - z\bar{w} - \bar{z}w + w\bar{w}$$

$$= 9 - (z\bar{w} + w\bar{z}) + 1 \quad (1)$$

$$= 10 - (z\bar{w} + w\bar{z})$$

$$= \text{RHS.}$$

(dy) $P(x) = 2x^3 + 3x^2 - x + 1$

roots are $\alpha^2, \beta^2, \gamma^2$

$$y^2 = x \quad \text{where } x = \alpha^2, \beta^2, \gamma^2$$

$$\therefore y = \sqrt{x}$$

$$\therefore [Q(x)] = 2(\sqrt{x})^3 + 3(\sqrt{x})^2 - \sqrt{x} + 1 = 0$$

$$2(\sqrt{x})^3 - \sqrt{x} = -3x - 1 \quad (1)$$

$$\sqrt{x}(2x-1) = -(3x+1)$$

$$x(4x^2 - 4x + 1) = (3x+1)^2$$

$$4x^3 - 4x^2 + x = 9x^2 + 6x + 1$$

$$\therefore 4x^3 - 13x^2 - 5x - 1 = 0$$

$$\therefore Q(x) = 4x^3 - 13x^2 - 5x - 1 \quad (1)$$

mod 2 $\alpha + \beta c$ $\rightarrow a+bi \rightarrow c+di$ $\sqrt{a^2+b^2}=3$
 $a^2+b^2=9$

$$\text{LHS} = |z-w|^2 = (a-c)^2 + (b-d)^2 = a^2+c^2-2ac+b^2+d^2-2bd$$

$$= 9+1-2(ac+bd) \quad (1)$$

$$\text{RHS} = 10 - [(a+bi)(c-di) + (a-bi)(c+di)]$$

$$= 10 - [ac+bd+bc i - adi + ac+bd - cb i + adi]$$

$$= 10 - [2ac+2bd]$$

$$= 10 - 2(ac+bd) \quad (1)$$

$$= \text{LHS}$$

method 3

$3\cos\theta + 3i\sin\theta$ $\cos\alpha + i\sin\alpha$

let $z = 3\text{cis}\theta$ $w = 1\text{cis}\alpha$

$$\text{LHS} = |z-w|^2 = (3\cos\theta - \cos\alpha)^2 + (3\sin\theta - \sin\alpha)^2$$

$$= 9\cos^2\theta + \cos^2\alpha - 6\cos\theta\cos\alpha + 9\sin^2\theta + \sin^2\alpha - 6\sin\theta\sin\alpha$$

$$= 9+1-6(\cos\theta\cos\alpha + \sin\theta\sin\alpha)$$

$$= 10 - 6(\cos\theta\cos\alpha + \sin\theta\sin\alpha) \quad (1)$$

$$\text{RHS} = 10 - [(3\cos\theta + 3i\sin\theta)(\cos\alpha - i\sin\alpha) + (3\cos\theta - 3i\sin\theta)(\cos\alpha + i\sin\alpha)]$$

$$= 10 - [3\cos\theta\cos\alpha - i3\cos\theta\sin\alpha + 3i\sin\theta\cos\alpha + 3\sin\theta\sin\alpha$$

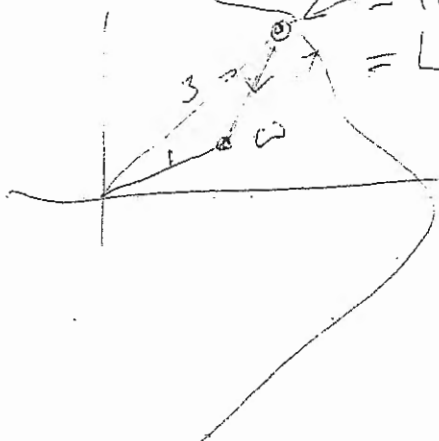
$$+ 3\cos\theta\cos\alpha + 3i\cos\theta\sin\alpha - 3i\sin\theta\cos\alpha + 3\sin\theta\sin\alpha]$$

method 4

$$= 10 - (6\cos\theta\cos\alpha + 6\sin\theta\sin\alpha)$$

$$= 10 - 6(\cos\theta\cos\alpha + \sin\theta\sin\alpha) \quad (1)$$

$$= \text{LHS}$$



(i)
$$\begin{cases} P(\alpha) = 2\alpha^3 + 3\alpha^2 - \alpha + 1 = 0 \\ P(\beta) = 2\beta^3 + 3\beta^2 - \beta + 1 = 0 \\ P(\gamma) = 2\gamma^3 + 3\gamma^2 - \gamma + 1 = 0 \end{cases}$$

$$2(\alpha^3 + \beta^3 + \gamma^3) = -3(\alpha^2 + \beta^2 + \gamma^2) + (\alpha + \beta + \gamma) - 3 \quad (1)$$

$$\alpha^2 + \beta^2 + \gamma^2 = -b/a = \frac{13}{4} \quad (\text{from part (i)})$$

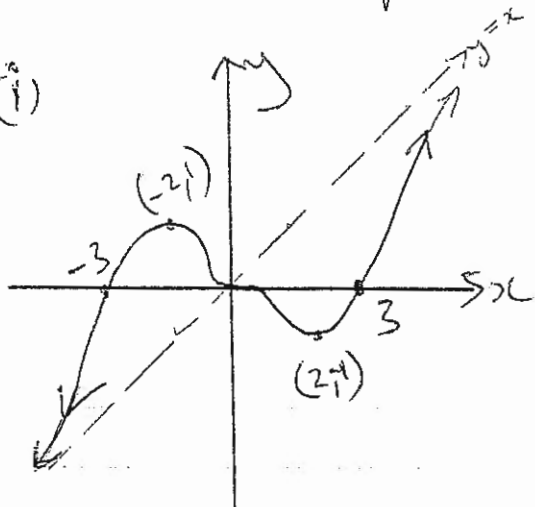
$$\alpha + \beta + \gamma = -3/2$$

$$\therefore 2(\alpha^3 + \beta^3 + \gamma^3) = -3 \times \frac{13}{4} + \frac{-3}{2} - 3$$

$$2(\alpha^3 + \beta^3 + \gamma^3) = -14\frac{1}{4}$$

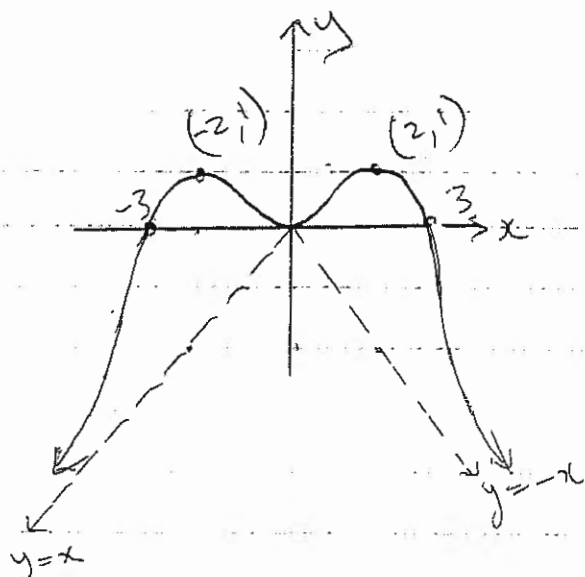
$$\alpha^3 + \beta^3 + \gamma^3 = -\frac{57}{8} \quad (1)$$

(e) (i)



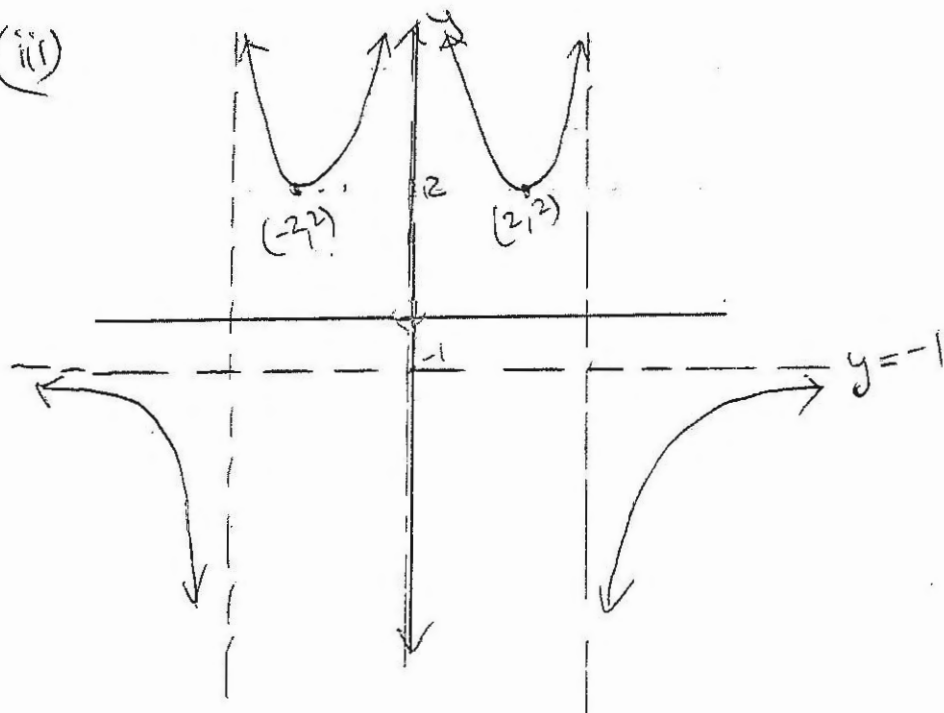
(1)

(ii)



(1)

(ii)



① for both branches above the x -axis

① for both branches below x -axis

① for $y = -1$, and turning pts $(\pm 2, 2)$ and a at $x = 0$

Question 13

(a) $P(x) = x^4 - x^3 - 2x^2 + 6x - 4$

as $(1+i)$ is a zero so is $(1-i)$
(all coefficients are rational)

$\therefore (x - (1+i))(x - (1-i))$ is a factor
ie $x^2 - 2x + 2$ is a factor

①

* now can either do a poly. division or use factor theorem.

$$P(-2) = 16 + 8 - 8 - 12 - 4 = 0$$

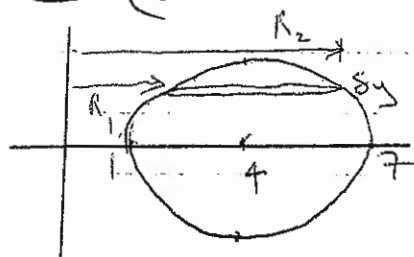
$\therefore (x+2)$ is a factor

$$P(1) = 1 - 1 - 2 + 6 - 4 = 0$$

$\therefore (x-1)$ is a factor

$$\begin{aligned} \therefore P(x) &= (x+2)(x-1)(x^2 - 2x + 2) \\ &= (x+2)(x-1)(x-1-i)(x-1+i) \end{aligned} \quad \text{①}$$

(b) (i)



$$\delta V = (\pi R_2^2 - \pi R_1^2) \delta y$$

$$x = 4 \pm \sqrt{1 - \frac{y^2}{4}}$$

$$R_2 + R_1 = 8 \quad R_2 - R_1 = 6\sqrt{1 - \frac{y^2}{4}}$$

$$\therefore \delta V = \pi (R_2^2 - R_1^2) \delta y$$
$$= \pi (R_2 - R_1)(R_2 + R_1) \delta y$$

①

$$= \pi \times 6\sqrt{1 - \frac{y^2}{4}} \times 8 \delta y$$

$$= \pi \frac{48}{2} \sqrt{4 - y^2} \delta y$$

①

$$= 8\pi \sqrt{36 - 9y^2}$$

(ii) $V = \lim_{\delta y \rightarrow 0} \sum_{-2}^2 8\pi \sqrt{36 - 9y^2} \delta y$

$$= 8\pi \int_{-2}^2 \sqrt{36 - 9y^2} dy$$

$$= 24\pi \int_{-2}^2 \sqrt{4 - y^2} dy$$

①

$$= 24\pi \times \frac{1}{2} (\pi \times 2^2)$$

(as $\sqrt{4 - y^2}$ is a semi-circle)

$$= 48\pi^2 \text{ units}^3$$

①

(c) $18x^3 + 3x^2 - 28x + 12 = 0$

let $P(x) = 18x^3 + 3x^2 - 28x + 12$

$$P'(x) = 54x^2 + 6x - 28$$

$$= 2(27x^2 + 3x - 14)$$

$$= 2(9x + 7)(3x - 2)$$

$$P'(x) = 0 \text{ when } x = -7/9 \text{ or } 2/3$$

①

$$P(-7/9) = 18(-7/9)^3 + 3(-7/9)^2 - 28(-7/9) + 12 \neq 0$$

$$\begin{aligned}
 P\left(\frac{2}{3}\right) &= 18\left(\frac{2}{3}\right)^3 + 3 \times \frac{4}{9} - \frac{56}{3} + 12 \\
 &= \frac{16}{3} + \frac{12}{9} - \frac{56}{3} + 12 \\
 &= 0
 \end{aligned}$$

$\therefore (3x-2)$ is a double root ①

$$\begin{aligned}
 \text{i.e. } P(x) &= (3x-2)^2(ax+\beta) \\
 &= (9x^2-12x+4)(ax+\beta)
 \end{aligned}$$

by inspection $a=2$, $\beta=3$

$$\therefore P(x) = (3x-2)^2(2x+3)$$

$\therefore$ roots are $\frac{2}{3}, \frac{2}{3}, -\frac{3}{2}$ ①

(d) (i) $P\left(5p, \frac{5}{p}\right) \quad Q\left(5q, \frac{5}{q}\right)$

$$m = \frac{\frac{5}{q} - \frac{5}{p}}{5q - 5p} = \frac{5\left(\frac{p-q}{pq}\right)}{5(q-p)} = \frac{-1}{pq} \quad \text{①}$$

eqn of PQ is

$$y - \frac{5}{p} = \frac{-1}{pq}(x - 5p)$$

$$pqy - 5q = -x + 5p \quad \text{①}$$

$$x + pqy = 5p + 5q$$

$$x + pqy = 5(p+q)$$

(ii) length of PQ is k^2 (data)

$$k^2 = \sqrt{5^2(p-q)^2 + 5^2\left(\frac{1}{p} - \frac{1}{q}\right)^2}$$

$$m^4 = 5^2 (p-q)^2 + 5^2 \left(\frac{1}{p} - \frac{1}{q}\right)^2 \quad (1)$$

$$m^4 = 5^2 (p-q)^2 + 5^2 \frac{(-p+q)^2}{p^2 q^2}$$

$$m^4 = 5^2 (p-q)^2 + \frac{5^2}{p^2 q^2} (p-q)^2 (-1)^2$$

$$m^4 = 25 (p-q)^2 \left[1 + \frac{1}{p^2 q^2}\right]$$

$$p^2 q^2 m^4 = 25 (p-q)^2 (p^2 q^2 + 1)$$

$$p^2 q^2 m^4 = 25 (p^2 + q^2 - 2pq)(p^2 q^2 + 1)$$

$$p^2 q^2 m^4 = 25 [(p+q)^2 - 4pq](p^2 q^2 + 1)$$

(ii) Midpoint of PQ is

$$M = \left(\frac{5p+5q}{2}, \frac{5/p+5/q}{2} \right)$$

$$\therefore R = \left(\frac{5(p+q)}{2}, \frac{5(\frac{p+q}{2pq})}{2} \right) \quad (R=M)$$

$$\therefore X = \frac{5}{2} (p+q) \quad Y = \frac{5}{2} (p+q) \frac{1}{pq} \quad (1)$$

$$\therefore p+q = \frac{2X}{5} \quad (1) \quad Y = \frac{X}{pq}$$

sub (1) and (2) into (ii)'s result. $pq = \frac{X}{Y} \quad (2)$

$$\therefore \frac{x^2}{y^2} m^4 = 25 \left[\frac{4x^2}{5^2} - \frac{4x}{y} \right] \left(\frac{x^2}{y^2} + 1 \right)$$

$$\frac{x^2}{y^2} m^4 = (4x^2 - \frac{4x \cdot 25}{y}) \left(\frac{x^2}{y^2} + 1 \right)$$

(now multiply both sides by y^2).

$$x^2 y m^4 = (4x^2 y - 4x \cdot 25) (x^2 + y^2)$$

$$2xy m^4 = 4(xy - 25)(x^2 + y^2)$$

(1)

Question 14

(a) (i) $P(4\cos\theta, 3\sin\theta)$

$$\frac{dx}{d\theta} = -4\sin\theta \quad \frac{dy}{d\theta} = 3\cos\theta$$

$$\frac{dy}{dx} = \frac{dy}{d\theta} \times \frac{d\theta}{dx}$$

$$= \frac{3\cos\theta}{-4\sin\theta}$$

(1)

eqn of tangent at P is $y - 3\sin\theta = \frac{-3\cos\theta}{4\sin\theta}(x - 4\cos\theta)$

$$y - 3\sin\theta = -\frac{3\cos\theta}{4\sin\theta}x + \frac{3\cos^2\theta}{\sin\theta}$$

$$\frac{3\cos\theta}{4\sin\theta}x + y = \frac{3\cos^2\theta}{\sin\theta} + 3\sin\theta$$

$$\frac{3\cos\theta}{4\sin\theta}x + y = \frac{3\cos^2\theta + 3\sin^2\theta}{\sin\theta}$$

$$\frac{3\cos\theta}{4\sin\theta}x + y = \frac{3}{\sin\theta}$$

(1)

$$\frac{\cos\theta}{4}x + \frac{y\sin\theta}{3} = 1 \dots (1)$$

At Q; $(4\cos\theta, 4\sin\theta)$

$$\frac{dx}{d\theta} = \frac{dy}{d\theta} \times \frac{d\theta}{dx}$$

$$= \frac{4\cos\theta}{-4\sin\theta} = -\frac{\cos\theta}{\sin\theta} \quad (1)$$

eqn of tangent at Q is $y - 4\sin\theta = -\frac{\cos\theta}{\sin\theta}(x - 4\cos\theta)$

$$y + \frac{x\cos\theta}{\sin\theta} = \frac{4\cos^2\theta}{\sin\theta} + \frac{4\sin^2\theta}{\sin\theta}$$

$$y + \frac{x\cos\theta}{\sin\theta} = \frac{4}{\sin\theta}$$

$$(1) \quad \frac{\sin\theta}{4}y + \frac{x\cos\theta}{4} = 1 \dots (2)$$

(ii) solve ① and ② simultaneously

$$y = \frac{3}{\sin \theta} - \frac{3 \cos \theta x}{4 \sin \theta} \quad \text{sub into ②}$$

$$\frac{3}{\sin \theta} - \frac{3 \cos \theta x}{4 \sin \theta} + \frac{x \cos \theta}{\sin \theta} = \frac{4}{\sin \theta}$$

$$3 - \frac{3}{4} \cos \theta x + x \cos \theta = 4$$

$$x \cos \theta \left(-\frac{3}{4} + 1 \right) = 1$$

$$x = \frac{4}{\cos \theta}$$

①

Sub. back

$$\therefore y = \frac{3}{\sin \theta} - \frac{3 \cos \theta}{4 \sin \theta} \cdot \frac{4}{\cos \theta}$$

$$y = \frac{3}{\sin \theta} - \frac{3}{\sin \theta}$$

$$y = 0$$

$$\therefore R \left(\frac{4}{\cos \theta}, 0 \right)$$

①

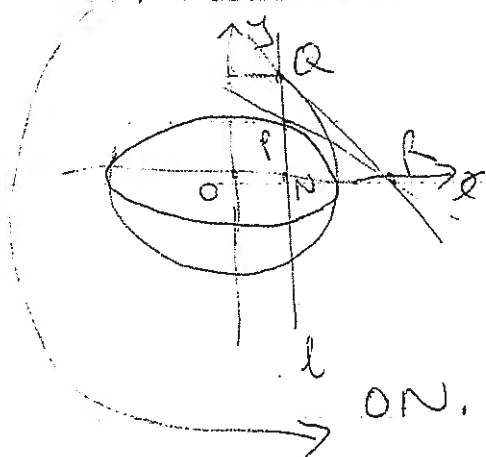
so R lies on the x-axis.

(iii)

ON · OR

$$N(4 \cos \theta, 0)$$

$$R \left(\frac{4}{\cos \theta}, 0 \right)$$



$$ON \cdot OR = 4 \cos \theta \times \frac{4}{\cos \theta}$$

①

ON · OR = 16 which is independent of P and Q.

(b) (i) $\cot 3\theta = -1$

$\therefore \tan 3\theta = -1$

$3\theta = \frac{3\pi}{4}, \frac{7\pi}{4}, \frac{11\pi}{4}, \frac{15\pi}{4}, \frac{19\pi}{4}, \frac{23\pi}{4}$

$\theta = \frac{\pi}{4}, \frac{7\pi}{12}, \frac{11\pi}{12}, \frac{15\pi}{12}, \frac{19\pi}{12}, \frac{23\pi}{12}$

(1) (1)

(ii) $\cot 3\theta = -1$ and $\cot 3\theta = \frac{t^3 - 3t}{3t^2 - 1}$

$\therefore -1 = \frac{t^3 - 3t}{3t^2 - 1}$

$1 - 3t^2 = t^3 - 3t$

$0 = t^3 + 3t^2 - 3t - 1$

but $t = \cot \theta$ (data)

$\therefore$ solns for t are $\cot \frac{\pi}{4}, \cot \frac{7\pi}{12}, \cot \frac{11\pi}{12}$

$(= \cot \frac{5\pi}{12})$
 $(= \cot \frac{2\pi}{3})$

(1) product of the roots is 0
 $\cot \frac{\pi}{4} \cdot \cot \frac{7\pi}{12} \cdot \cot \frac{11\pi}{12} = -d/a = 1$

$\cot \frac{\pi}{4} \cdot \cot \frac{7\pi}{12} \cdot \cot \frac{11\pi}{12} = 1 \dots \dots \textcircled{1}$

now $\cot(-\theta) = -\cot \theta$ (odd fn)
deduct 2π from each of the angles in (1)

$\therefore -\cot \frac{\pi}{4} \times -\cot \frac{5\pi}{12} \times -\cot \frac{9\pi}{12} = 1$

(1)

$-1 \times \cot \frac{\pi}{4} \times \cot \frac{5\pi}{12} \times \cot \frac{9\pi}{12} = 1$

$\therefore \cot \frac{\pi}{4} \cdot \cot \frac{5\pi}{12} \cdot \cot \frac{9\pi}{12} = -1$

$\therefore \cot \left(\frac{\pi}{4}\right) \cdot \cot \left(\frac{5\pi}{12}\right) \cdot \cot \left(\frac{9\pi}{12}\right)$

(ii) eqn with roots $\tan \frac{\pi}{12}$, $\tan \frac{5\pi}{12}$, $\tan \frac{9\pi}{12}$

ie $\frac{1}{\cot \frac{\pi}{12}}$, $\frac{1}{\cot \frac{5\pi}{12}}$, $\frac{1}{\cot \frac{9\pi}{12}}$

ie $\frac{1}{t}$

①

$\therefore$ eqn is $\left(\frac{1}{t}\right)^3 + 3\left(\frac{1}{t}\right)^2 - 3\left(\frac{1}{t}\right) - 1 = 0$

$1 + 3t - 3t^2 - t^3 = 0$ ①

$\therefore [t^3 + 3t^2 - 3t - 1 = 0]$

(c) (i) $y = x^x$ for $x > 0$

$y = e^{x \ln x}$ $\leftarrow \begin{matrix} \Rightarrow \ln y = \ln x^x \\ \ln y = x \ln x \end{matrix}\right.$

$\frac{dy}{dx} = e^{x \ln x} \times \left[x \times \frac{1}{x} + \ln x \times 1 \right]$

$= e^{x \ln x} (1 + \ln x)$

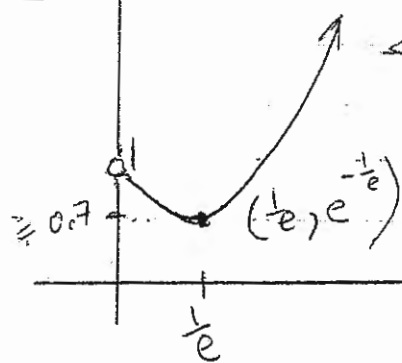
$= e^{\ln x^x} (1 + \ln x)$

$= x^x (1 + \ln x)$

①

(ii)

$\frac{dy}{dx} = 0$ when $\ln x = -1$ $x = e^{-1}$

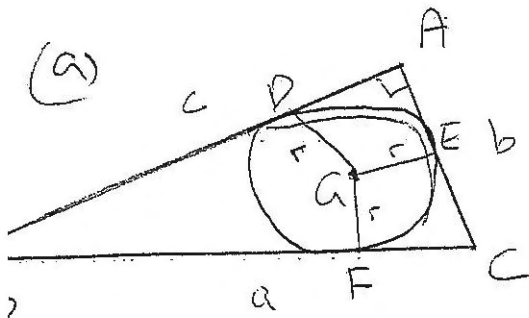


$\leftarrow$ should be quite steep

① for shape.

① for min. turning point and starting at $(0, 1)$

Question 15



$$\hat{ADG} = 90^\circ \text{ (tangent meets radius at right angles)}$$

$$\text{Similarly } \hat{AEG} = 90^\circ \text{ and } \hat{GFC} = 90^\circ$$

$$\hat{A} = 90^\circ \text{ (data)}$$

$$AD = AE \text{ (tangents from external point are equal.)}$$

In $\triangle AGE$,
all angles are 90° (proven above)
 $AD = AE$ (proven above)

$\therefore ADGE$ is a square

$$\therefore AD = AE = DG = GE = r$$

$$\therefore BD = c - r$$

$$BD = BF \text{ (tangents drawn from external point)}$$

$$\therefore BF = c - r \text{ (as } BD = c - r)$$

$$\text{now } AE = r$$

$$\therefore EC = b - r$$

$$CF = b - r \text{ (as } CF = EC)$$

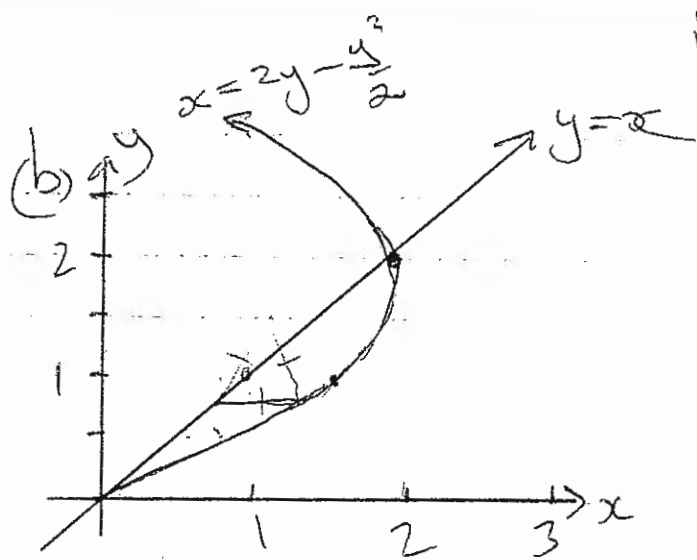
$$\therefore BC = BF + FC \text{ (sum of adjacent lengths)}$$

$$a = c - r + b - r$$

$$a = c + b - 2r$$

$$a - c - b = -2r$$

$$\therefore r = \frac{1}{2}(b + c - a)$$

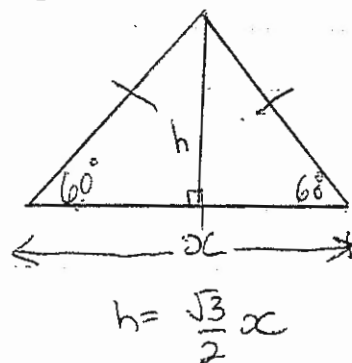


intercepts:

$$y = 2y - \frac{y^2}{2}$$

$$y^2 - 2y = 0$$

$$y = 0 \text{ or } y = 2$$



$$h = \frac{\sqrt{3}}{2} x$$

length of side of triangle $= x_2 - x_1$

$$= 2y - \frac{y^2}{2} - y$$

$$= y - \frac{y^2}{2}$$

①

Area of triangle $= \frac{1}{2} ab \sin 60^\circ$

$$= \frac{1}{2} x^2 \frac{\sqrt{3}}{2}$$

$$= \frac{\sqrt{3}}{4} \left(y - \frac{y^2}{2} \right)^2$$

①

Volume of single triangular prism:

$$\delta V = \frac{\sqrt{3}}{4} \left(y - \frac{y^2}{2} \right)^2 \delta y$$

$\therefore$ Volume of solid:

$$V = \lim_{\delta y \rightarrow 0} \sum_{y=0}^2 \frac{\sqrt{3}}{4} \left(y - \frac{y^2}{2} \right)^2 \delta y$$

$$= \frac{\sqrt{3}}{4} \int_0^2 \left(y - \frac{y^2}{2} \right)^2 dy$$

①

$$= \frac{\sqrt{3}}{4} \int_0^2 \left(y^2 + \frac{y^4}{4} - y^3 \right) dy$$

$$(ii) \quad = \frac{\sqrt{3}}{4} \left[\frac{1}{3} y^3 + \frac{y^5}{20} - \frac{1}{4} y^4 \right]_0^2 \quad (1)$$

$$= \frac{\sqrt{3}}{4} \left[\frac{1}{3} \times 8 + \frac{32}{20} - \frac{16}{4} - 0 \right]$$

$$= \frac{\sqrt{3}}{4} \left[\frac{160}{60} + \frac{96}{60} - \frac{240}{60} \right] = \frac{\sqrt{3}}{4} \times \frac{16}{60}$$

$$= \frac{\sqrt{3}}{15} \text{ units}^3 \quad (1)$$

(c) (i)

$$I_n = \int_0^1 (1-x^2)^{n/2} dx$$

$$= \left[x(1-x^2)^{n/2} \right]_0^1 + n \int_0^1 x^2 (1-x^2)^{n/2-1} dx \quad (1)$$

$$= 0 + n \int_0^1 (1-x^2+1)(1-x^2)^{n/2-1} dx$$

$$= -n \int_0^1 (1-x^2)^{\frac{n}{2}} dx + \int_0^1 (1-x^2)^{n/2-1} dx \quad (1)$$

$$= -n I_n + n I_{n-2} \quad \left(\text{as } \frac{n}{2}-1 = \frac{n-2}{2} \right) \quad (1)$$

$$I_n + n I_n = n I_{n-2}$$

$$I_n (1+n) = n I_{n-2}$$

$$I_n = \frac{n}{n+1} I_{n-2}$$

(c) (ii) Evaluate I_5

$$I_5 = \frac{5}{6} I_3$$

$$I_3 = \frac{3}{4} I_1$$

$$I_1 = \int_0^1 \sqrt{1-x^2} dx$$

$\frac{1}{4}$ of a circle

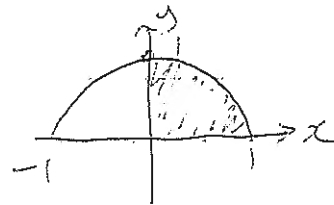
$$\therefore I_1 = \frac{1}{4} \pi \times 1^2 = \frac{\pi}{4}$$

①

$$\therefore I_3 = \frac{3}{4} \times \frac{\pi}{4} = \frac{3\pi}{16}$$

$$I_5 = \frac{5}{6} \times \frac{3\pi}{16} = \frac{5\pi}{32}$$

①



(d) $(2-m^2) > 0$

$$m^2 < 2$$
$$-\sqrt{2} < m < \sqrt{2}$$

but $m > 0$ given

①

$$\therefore 0 < m < \sqrt{2}$$

①

Q16(a)(i) $a > 0, b > 0$ given $a \neq b$

now $(a-b)^2 \geq 0$

$$a^2 + b^2 - 2ab \geq 0$$

$$a^2 + b^2 \geq 2ab$$

①

(ii)

now $a^2 + b^2 \geq 2ab$

similarly $a^2 + c^2 \geq 2ac$

$b^2 + c^2 \geq 2bc$

$$\therefore 2(a^2 + b^2 + c^2) \geq 2(ab + ac + bc) \quad (1)$$

$$a^2 + b^2 + c^2 \geq ab + ac + bc$$

(iii) Let $a = \sin x$ $b = \cos x$

now $a^2 + b^2 \geq 2ab$ from (i)

$$\sin^2 x + \cos^2 x \geq 2 \sin x \cos x \quad (1)$$

$$\therefore \sin^2 x + \cos^2 x \geq \sin 2x$$

(iv) $a^2 + b^2 + c^2 \geq ab + ac + bc$ from (ii)

let $a = \sin x$, $b = \cos x$, $c = \tan x$

$$\therefore \sin^2 x + \cos^2 x + \tan^2 x \geq \sin x \cos x + \cos x \tan x + \sin x \tan x$$

$$\begin{aligned} &\geq \frac{1}{2} \sin 2x + \sin x + \frac{\sin^2 x}{\cos x} \\ &= \frac{1}{2} \sin 2x + \sin x + \frac{1 - \cos^2 x}{\cos x} \\ &= \frac{1}{2} \sin 2x + \sin x + \sec x - \cos x \end{aligned} \quad (1)$$

(b) Paul has been chosen, 2 females

so committee is 1m, 2w

$$\begin{aligned} \text{Prob} &= \frac{{}^{n-1}C_1}{{}^nC_2} \\ &= \frac{(n-1)!}{(n-2)!} \times \frac{2!(n-2)!}{n!} \\ &= \frac{2}{n} \end{aligned} \quad (1)$$

$$(c) (i) \quad P(x) = (x^2 - a^2)Q(x) + px + q$$

$$= (x-a)(x+a)Q(x) + px + q$$

$$P(a) = pa + q \quad \dots \quad (1)$$

$$P(-a) = -ap + q \quad \dots \quad (2)$$

$$(1) - (2)$$

$$P(a) - P(-a) = pa + q - (-ap + q) = 2ap$$

(1)

$$\therefore p = \frac{1}{2a} [P(a) - P(-a)]$$

$$(1) + (2)$$

$$P(a) + P(-a) = pa + q - ap + q = 2q$$

(1)

$$\therefore q = \frac{1}{2} [P(a) + P(-a)]$$

$$ii) \text{ Given } P(x) = x^n - a^n$$

$$\text{if } n \text{ is even; } P(a) = a^n - a^n = 0$$

$$P(-a) = (-a)^n - a^n = 0$$

$$\text{from (i) remainder is } p + q$$

$$p = 0, q = 0$$

$$\therefore \text{no remainder}$$

(1)

$$\text{if } n \text{ is odd; } P(a) = a^n - a^n = 0$$

$$P(-a) = (-a)^n - a^n = -a^n - a^n = -2a^n$$

$$p = \frac{1}{2a} [0 - (-2a^n)] = a^{n-1}$$

$$q = \frac{1}{2} [0 + (-2a^n)] = -a^n$$

$$\therefore P(x) = a^{n-1}x - a^n$$

(1)

$$d) (i) S_n = 1 + \sum_{r=1}^n \frac{1}{r!}$$

$$e - S_n = e \int_0^1 \frac{x^n}{n!} e^{-x} dx$$

step one: Prove true for $n=1$

$$\begin{aligned} \text{LHS} &= e - S_1 \\ &= e - \left(1 + \frac{1}{1!}\right) \\ &= e - 2 \end{aligned}$$

$$\begin{aligned} \text{RHS} &= e \int_0^1 \frac{x^1}{1!} e^{-x} dx \\ &= e \int_0^1 x e^{-x} dx \\ &= e \left[(-x e^{-x}) \right]_0^1 + \int_0^1 e^{-x} dx \\ &= e \left(-e^{-1} - 0 \right) + \left[-e^{-x} \right]_0^1 \\ &= e \left(-e^{-1} - e^{-1} + e^0 \right) \\ &= e \left(-2e^{-1} + 1 \right) \\ &= e \left(-2\frac{1}{e} + 1 \right) \\ &= e - 2 \\ &= \text{LHS.} \end{aligned} \quad (1)$$

step two: Assume true for $n=k$, k is an integer

$$\text{i.e. } e - S_k = e \int_0^1 \frac{x^k}{k!} e^{-x} dx$$

step three: Prove true for $n=k+1$,

$$\text{i.e. } e - S_{k+1} = e \int_0^1 \frac{x^{k+1}}{(k+1)!} e^{-x} dx$$

$$\text{LHS} = e - S_{k+1}$$

$$= e - \left(1 + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \dots + \frac{1}{k!} + \frac{1}{(k+1)!} \right)$$

$$= e \int_0^1 \frac{x^k}{k!} e^{-x} dx - \frac{1}{(k+1)!} \quad (\text{by assumption}) \quad (1)$$

$$\text{RHS} = e \int_0^1 \frac{x^{k+1}}{(k+1)!} e^{-x} dx$$

$$u = x^{k+1} \quad v = -e^{-x}$$

$$\frac{du}{dx} = (k+1)x^k \quad \frac{dv}{dx} = e^{-x}$$

$$= \frac{e}{(k+1)!} \left[-x^{k+1} e^{-x} \right]_0^1 - \frac{e}{(k+1)!} \int_0^1 (k+1)x^k e^{-x} dx$$

$$= \frac{e}{(k+1)!} x e^{-x} + \frac{e}{k!} \int_0^1 x^k e^{-x} dx \quad (1)$$

$$= \frac{-1}{(k+1)!} + e \int_0^1 \frac{x^k}{k!} e^{-x} dx$$

$$= \text{LHS.}$$

Step Four: By the Principle of Mathematical Induction, the result is true for all integers, $n \geq 1$.

$$(ii) \text{ Deduce } 0 < e - s_n < \frac{3}{(n+1)!}$$

$$\text{now } e \int_0^1 \frac{x^n}{n!} e^{-x} dx > 0 \text{ as } e > 0 \text{ and } e^{-x} > 0$$

$$\text{and } x^n \geq 0 \text{ for } x \geq 0$$

$$e < 3: e \int_0^1 \frac{x^n}{n!} e^{-x} dx < 3 \int_0^1 \frac{x^n}{n!} e^{-x} dx$$

$$< 3 \int_0^1 \frac{x^n}{n!} dx \text{ as } e^{-x} \leq 1$$

$$(1) = 3 \left[\frac{x^{n+1}}{n!(n+1)} \right]_0^1 \text{ for } 0 < x < 1$$

$$= \frac{3}{(n+1)!} - 0$$

$$\therefore 0 < e - s_n < \frac{3}{(n+1)!}$$

(iii) Prove that $(e - s_n)n!$ is not an integer

$$\text{now } e - s_n < \frac{3}{n!(n+1)} \quad \text{from (ii)}$$

$$(e - s_n)n! < \frac{3}{n+1}$$

$$\frac{3}{n+1} \leq 1 \quad \text{for } n = 2, 3, 4, \dots$$

$$\therefore (e - s_n)n! < 1 \quad \textcircled{1}$$

$\therefore (e - s_n)n!$ is not an integer for $n = 2, 3, 4, \dots$

(iv) Assume $e = \frac{p}{q}$

$$\text{now } (e - s_n)n! = \left[e - \left(1 + \sum_{r=1}^n \frac{1}{r!} \right) \right] n!$$

$$= \left(\frac{p}{q} - 1 - \sum_{r=1}^n \frac{1}{r!} \right) n!$$

$$= \frac{pn!}{q} - n! - \left(\sum_{r=1}^n \frac{1}{r!} \right) n!$$

now as q is any integer and $n! = n(n-1)(n-2) \times \dots \times 3 \times 2 \times 1$

which includes " q ", $\therefore n!$ is divisible by q

For $n > r$, then $n! \times \sum_{r=1}^n \frac{1}{r!}$ must be an integer

$$\therefore \frac{pn!}{q} - n! - \left(\sum_{r=1}^n \frac{1}{r!} \right) n! \text{ is an integer} \quad \textcircled{1}$$

but this contradicts (iii) where $(e - s_n)n!$ is not an integer. $\therefore e \neq \frac{p}{q}$ so e is not rational.